“SnoozeStats: Mapping Landscapes of Sleep & Lifestyle Variations” - Data Analysis

Basics of Dataset

In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

IMPORTING DATASET INTO PANDA'S DATAFRAME

In [3]:
```
df = pd.read_csv('/Sleep_health_and_lifestyle_dataset.csv')
```

PRINTING DATASET ALONG WITH THEIR NUMBER OF ROWS & COLUMNS

In [ ]:
```
df.shape
```

Out[ ]:
```
(374, 13)
```

In [ ]:
```
df
```

Out[ ]:

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	1	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	None
1	2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
2	3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
...	...	...	...	...	...	...	...	...	...	...	...	...	...
369	370	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
370	371	Female	59	Nurse	8.0	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
371	372	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
372	373	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea
373	374	Female	59	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	Sleep Apnea

374 rows × 13 columns

In [ ]:
```
df.head(5)
```

Out[ ]:

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	1	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	None
1	2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
2	3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
3	4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea

LOOKING FOR MISSING VALUES IN THE DATA SET

TRUE = CONTAINS MISSING/NULL VALUES

FALSE = DO NOT CONTAIN MISSING/NULL VALUES

In [ ]:
```
missing_value = df.isnull()
```

```
missing_value
```

Out[ ]:

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...	...
369	False	False	False	False	False	False	False	False	False	False	False	False	False
370	False	False	False	False	False	False	False	False	False	False	False	False	False
371	False	False	False	False	False	False	False	False	False	False	False	False	False
372	False	False	False	False	False	False	False	False	False	False	False	False	False
373	False	False	False	False	False	False	False	False	False	False	False	False	False

374 rows × 13 columns

In [ ]:

```
missing_count = missing_value.sum()
missing_count
```

Out[ ]:

```
Person ID                 0
Gender                    0
Age                       0
Occupation                0
Sleep Duration            0
Quality of Sleep          0
Physical Activity Level   0
Stress Level              0
BMI Category              0
Blood Pressure            0
Heart Rate                0
Daily Steps               0
Sleep Disorder            0
dtype: int64
```

Descriptive Statistics for DataFrame

In [ ]:

```
df.describe()
```

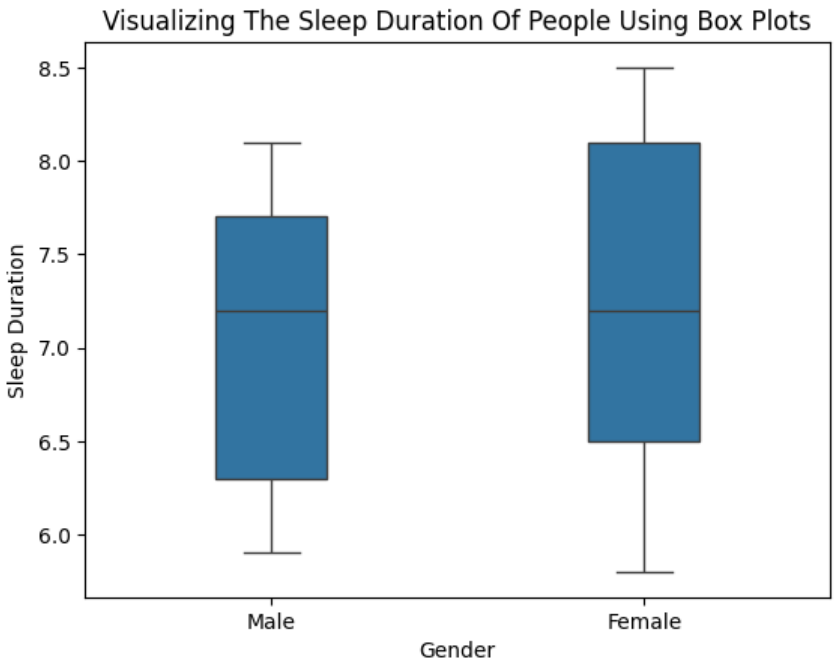
Out[ ]:

	Person ID	Age	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	Heart Rate	Daily Steps
count	374.000000	374.000000	374.000000	374.000000	374.000000	374.000000	374.000000	374.000000
mean	187.500000	42.184492	7.132086	7.312834	59.171123	5.385027	70.165775	6816.844920
std	108.108742	8.673133	0.795657	1.196956	20.830804	1.774526	4.135676	1617.915679
min	1.000000	27.000000	5.800000	4.000000	30.000000	3.000000	65.000000	3000.000000
25%	94.250000	35.250000	6.400000	6.000000	45.000000	4.000000	68.000000	5600.000000
50%	187.500000	43.000000	7.200000	7.000000	60.000000	5.000000	70.000000	7000.000000
75%	280.750000	50.000000	7.800000	8.000000	75.000000	7.000000	72.000000	8000.000000
max	374.000000	59.000000	8.500000	9.000000	90.000000	8.000000	86.000000	10000.000000

Distribution of Dataset

In [ ]:

```
sns.boxplot(x='Gender', y='Sleep Duration', data= df, width= 0.3, linewidth = 1)
plt.title('Visualizing The Sleep Duration Of People Using Box Plots')
plt.show()
```

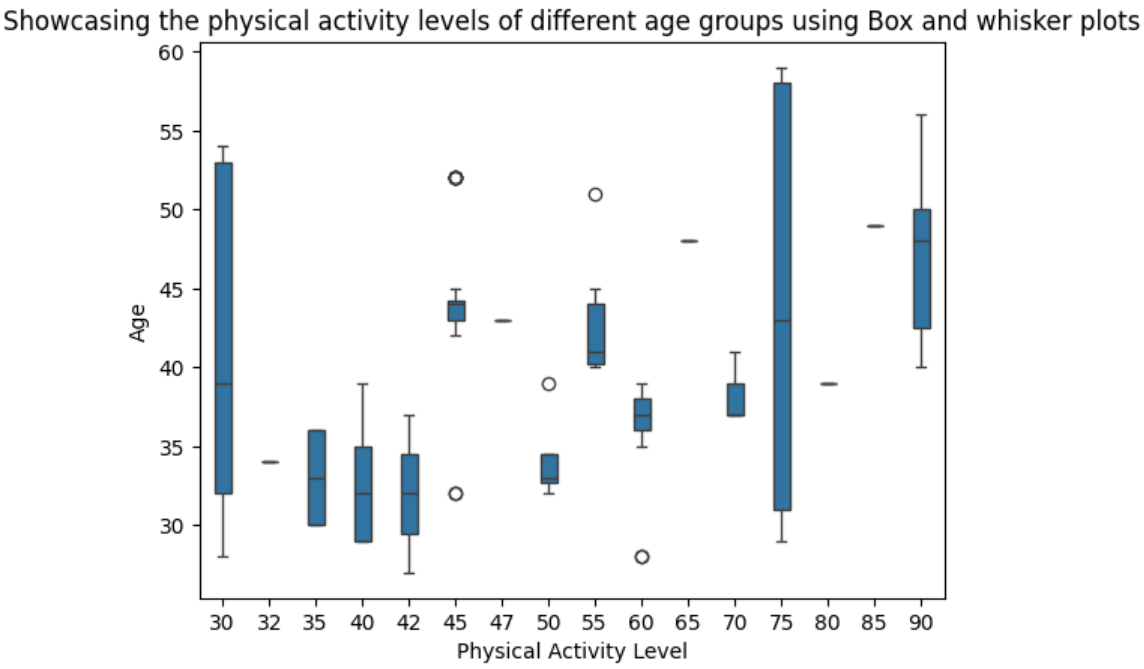


For males, the mean sleep duration is approximately 7.5 hours, with the majority falling within the range of 6 to 8 hours. The box plot indicates a tight distribution, suggesting consistency in sleep patterns. The upper whisker extends to 8 hours, indicating the maximum sleep duration observed, while the lower whisker reaches 6 hours, representing the minimum sleep duration.

In contrast, females exhibit a slightly lower mean sleep duration of around 7 hours. The box plot for females shows a wider distribution, ranging from 6 to 8.5 hours. The upper whisker extends to 8.5 hours, indicating the maximum observed sleep duration, while the lower whisker reaches 6 hours.

However, the analysis suggests that the variability in sleep duration is more pronounced among females, with some individuals experiencing both shorter and longer sleep durations.

In [ ]:
```
sns.boxplot(df, y='Age', x= 'Physical Activity Level', width=0.35)
plt.title('Showcasing the physical activity levels of different age groups using Box and whisker plots')
plt.show()
```

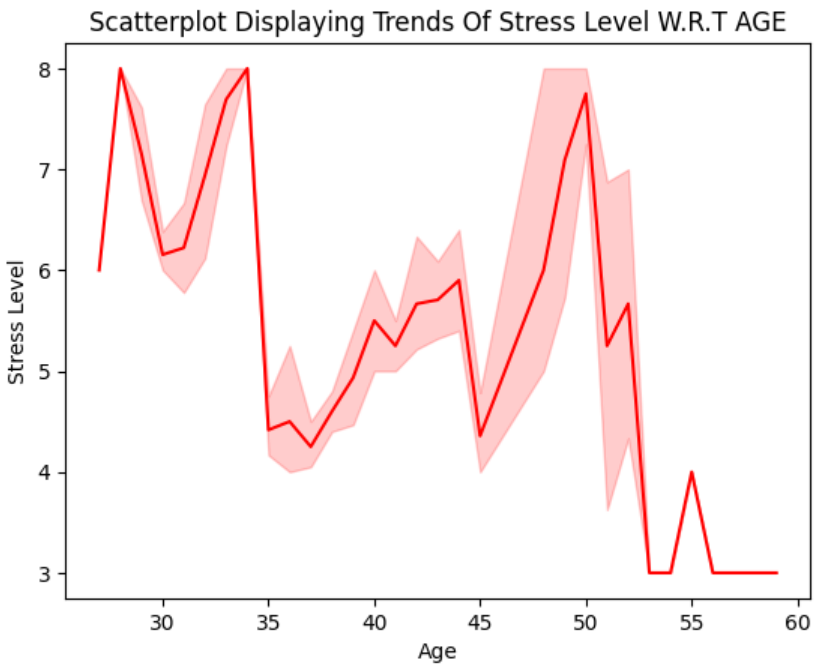


The mean age for most activity levels is concentrated within the range of 30 to 45 years. This suggests that individuals within this age group tend to have similar average physical activity levels.

The longest whisker is observed for activity levels 30 and 75. This indicates that these specific activity levels have a higher variability in ages. The spread of ages for these activity levels is wider, potentially suggesting a more diverse age distribution and varied physical activity behaviors.

For activity level 30 and 75, where the mean age is around 40, the presence of the longest whisker may indicate that individuals within these activity levels exhibit a broader age range.

In [9]:
```
sns.lineplot(df,x='Age', y='Stress Level', color='Red')
plt.title('Scatterplot Displaying Trends Of Stress Level W.R.T AGE')
plt.show()
```



The line plot reveals a noteworthy pattern in stress levels concerning age. Between the ages of 35 and 45, there is a decreasing trend, with stress levels declining from 8 to 6. Following this, between ages 45 and 55, stress levels remain consistently stable. This suggests that individuals experience a decrease in stress during their mid-thirties to mid-forties, followed by a period of sustained stress levels in the subsequent decade.

Correlation Matrix

In [ ]:
```
correlation_coefficients = df.corr()
correlation_data =df.corr()
correlation_data
```

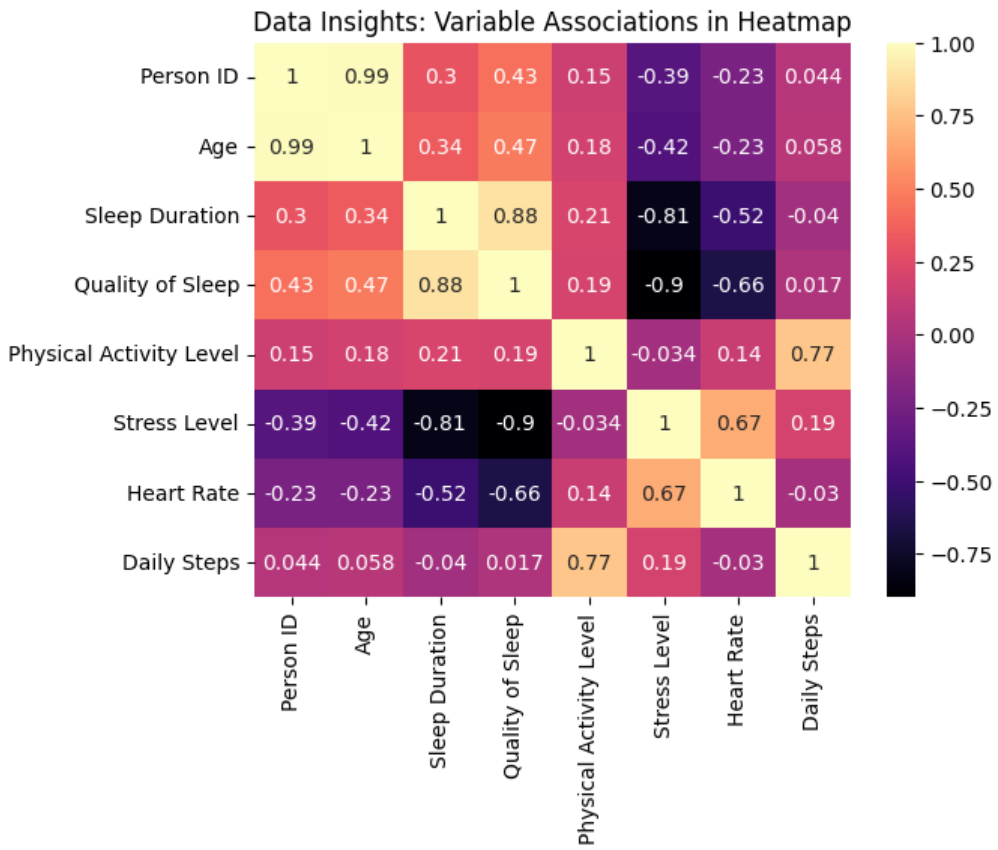
```
<ipython-input-35-6423134896a6>:1: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated. In a future version, it will default to Fa
lse. Select only valid columns or specify the value of numeric_only to silence this warning.
  correlation_coefficients = df.corr()
<ipython-input-35-6423134896a6>:2: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated. In a future version, it will default to Fa
lse. Select only valid columns or specify the value of numeric_only to silence this warning.
  correlation_data =df.corr()
```

Out[ ]:

	Person ID	Age	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	Heart Rate	Daily Steps
Person ID	1.000000	0.990516	0.296305	0.431612	0.149882	-0.394287	-0.225467	0.043844
Age	0.990516	1.000000	0.344709	0.473734	0.178993	-0.422344	-0.225606	0.057973
Sleep Duration	0.296305	0.344709	1.000000	0.883213	0.212360	-0.811023	-0.516455	-0.039533
Quality of Sleep	0.431612	0.473734	0.883213	1.000000	0.192896	-0.898752	-0.659865	0.016791
Physical Activity Level	0.149882	0.178993	0.212360	0.192896	1.000000	-0.034134	0.136971	0.772723
Stress Level	-0.394287	-0.422344	-0.811023	-0.898752	-0.034134	1.000000	0.670026	0.186829
Heart Rate	-0.225467	-0.225606	-0.516455	-0.659865	0.136971	0.670026	1.000000	-0.030309
Daily Steps	0.043844	0.057973	-0.039533	0.016791	0.772723	0.186829	-0.030309	1.000000

In [ ]:

```
sns.heatmap(correlation_data, annot=True, cmap="magma")
plt.title("Data Insights: Variable Associations in Heatmap")
plt.show()
```



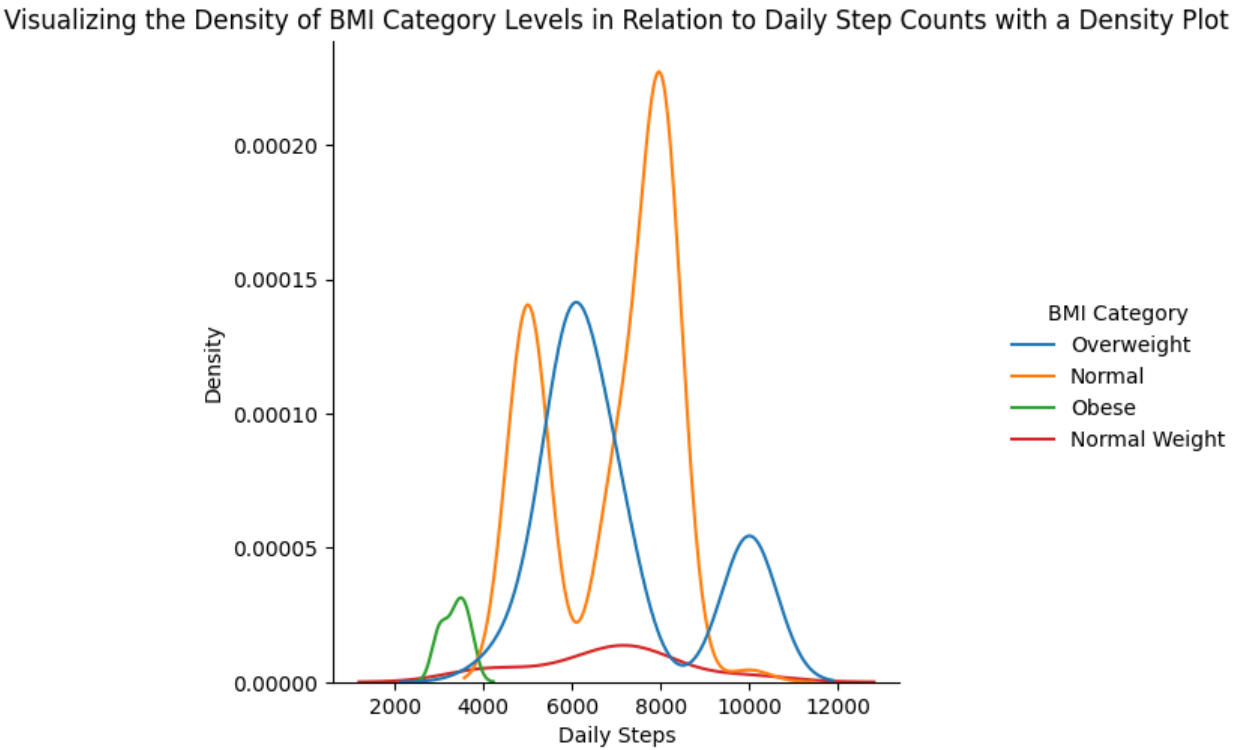
The high correlation coefficients are 0.88, 0.77 and 0.67 that suggests strong positive relationships. These relations are between Sleep Duration - Quality of Sleep, Daily Steps - Physical Activity Level and Stress Level - Heart Rate

A strong positive correlation of 0.88 between sleep duration and the quality of sleep indicates a robust relationship. This suggests that as sleep duration increases, the quality of sleep tends to improve, and vice versa. These variables are positively associated, meaning that individuals who sleep longer are more likely to report higher sleep quality.

The positive correlation indicates that as the number of daily steps increases, there tends to be a corresponding improvement in the physical quality of sleep, and vice versa.This finding may have significant implications for health and well-being, as it suggests a potential connection between physical activity and sleep quality. Individuals who engage in higher levels of daily steps may experience better physical sleep quality, which aligns with existing research highlighting the positive impact of physical activity on sleep patterns.

The positive correlation of 0.67 between stress level and heart rate implies a moderate association between these variables. This suggests that higher stress levels are moderately associated with higher heart rates. It's important to note that correlation does not imply causation, but in this case, the correlation suggests a connection between increased stress and elevated heart rates.

In [ ]:
```
sns.displot(df, x='Daily Steps', hue='BMI Category', kind='kde')
plt.title('Visualizing the Density of BMI Category Levels in Relation to Daily Step Counts with a Density Plot')
plt.show()
```



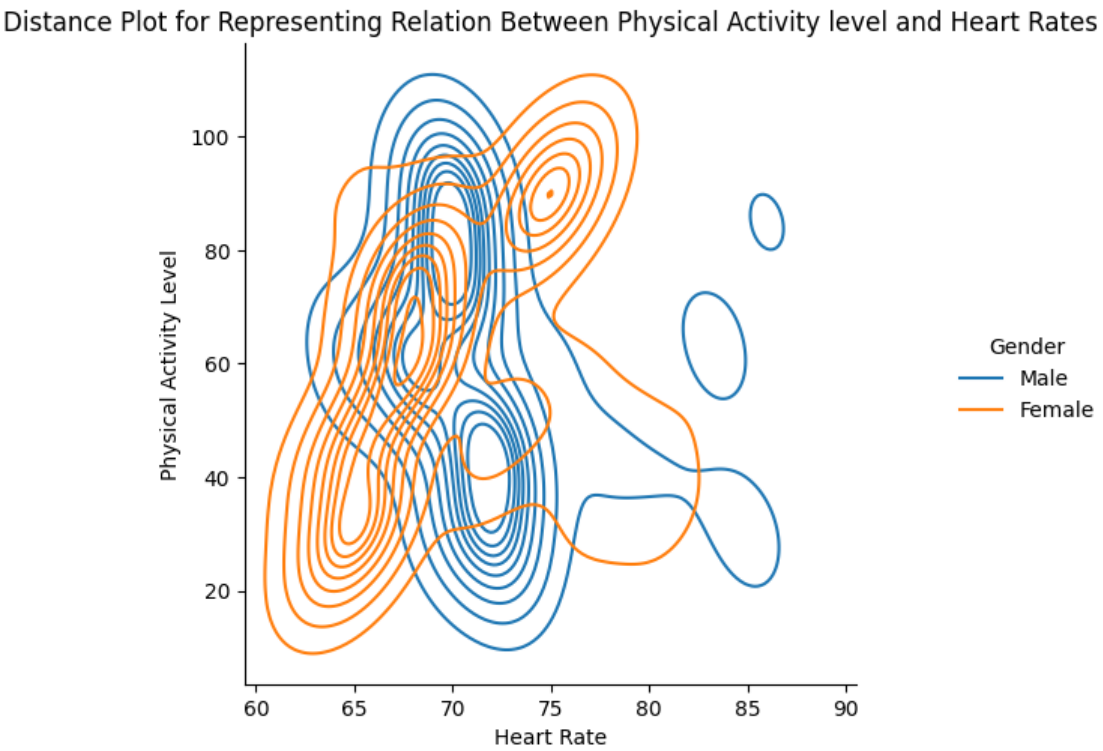
A concentration of daily steps between 6000 to 9000 being associated with the "Normal" BMI category suggests that individuals within this steps range are more likely to have a BMI within the normal weight range. This aligns with general health recommendations emphasizing the positive impact of moderate physical activity on maintaining a healthy weight.

The density plot indicates that daily steps falling between 4000 to 6000 are associated with the "Overweight" BMI category. This suggests that individuals with lower daily steps within this range may have an increased likelihood of being classified as overweight.

The observation that daily steps below 4000 are associated with the "Obese" BMI category implies that individuals with lower physical activity levels may be more prone to obesity. This finding underscores the importance of regular physical activity in weight management.

These associations have practical implications for public health interventions and individual well-being. Encouraging higher daily step counts may be a beneficial strategy for weight management and reducing the risk of obesity.

In [ ]:
```
sns.displot(df, x='Heart Rate', y='Physical Activity Level', hue= 'Gender', kind='kde')
plt.title('Distance Plot for Representing Relation Between Physical Activity level and Heart Rates')
plt.show()
```

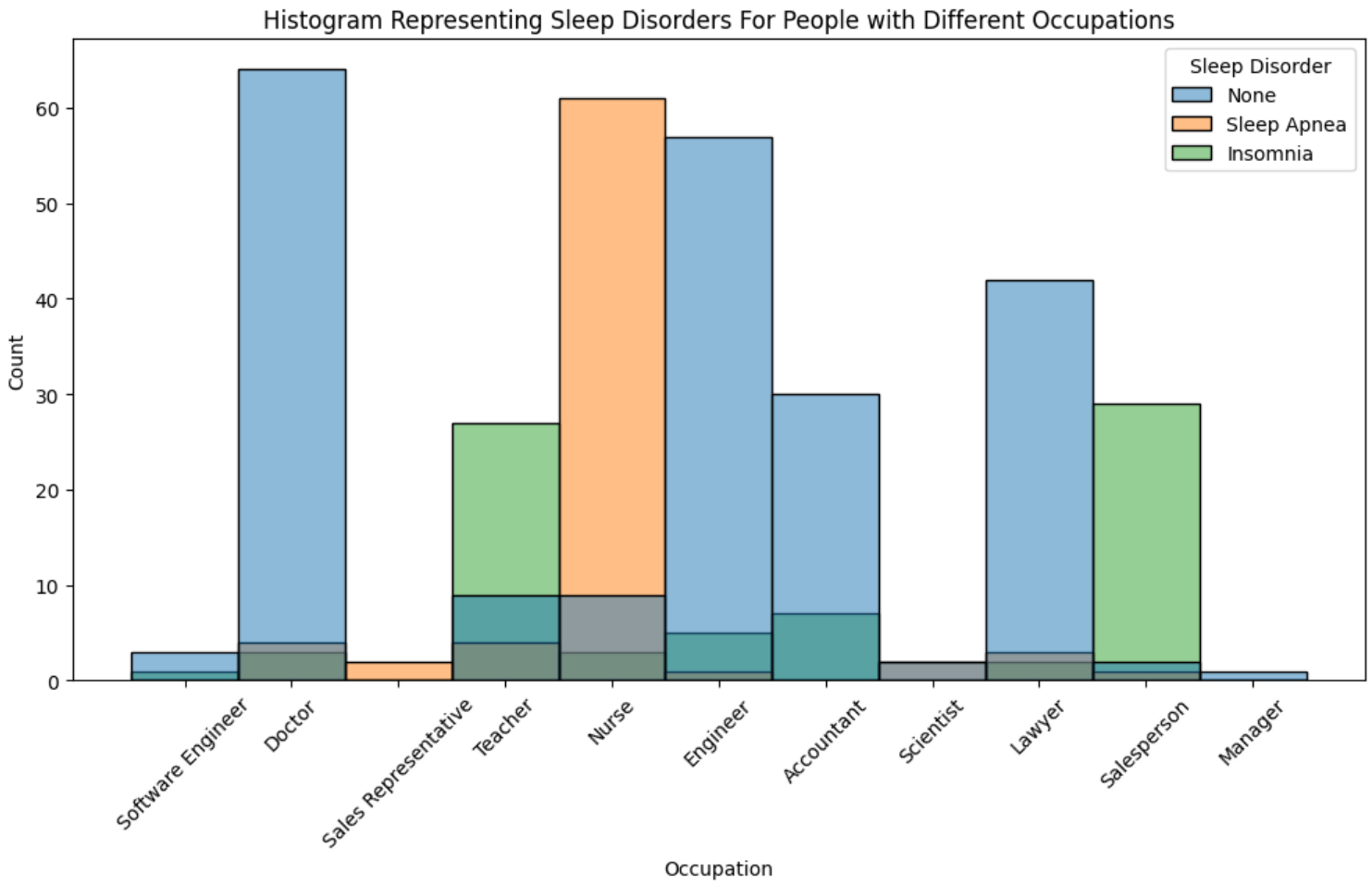


The observed overlap of clusters around heart rate 65 and physical activity level 70 between both genders implies that there is a common range of heart rate and physical activity level where individuals from both genders tend to cluster

This finding may suggest that individuals with heart rates around 70 and engaging in physical activity within the specified range may share common cardiovascular fitness levels. A heart rate of 70 is often considered within the normal range and may be reflective of efficient cardiovascular function.

The presence of outliers with heart rates between 80-90 for males and physical activity levels between 60-80 suggests that there are individuals within the male group who exhibit higher heart rates while engaging in moderate to high physical activity.

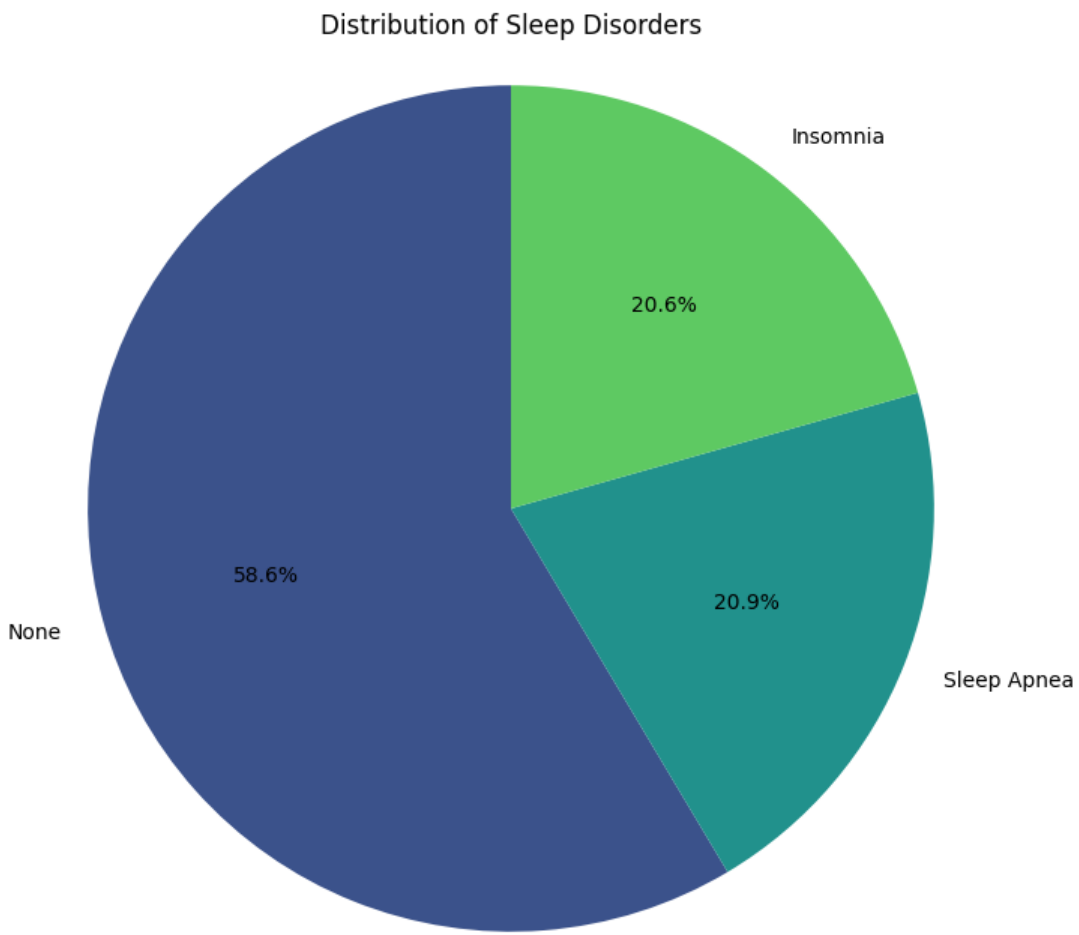
In [ ]:
```
plt.figure(figsize=(12, 6))
sns.histplot(df, x='Occupation', hue='Sleep Disorder')
plt.xticks(rotation=45)
plt.title('Histogram Representing Sleep Disorders For People with Different Occupations')
plt.show()
```



Most occupations show a low prevalence of sleep disorders, indicating that the majority of people surveyed do not report significant sleep-related issues.The histogram reveals that occupations such as Nurses, Teachers, and Salespeople stand out with a higher prevalence of sleep disorders. This could be attributed to the nature of these professions, which often involve irregular working hours, high stress levels, and increased responsibility.

Aggregation of Categorical Data

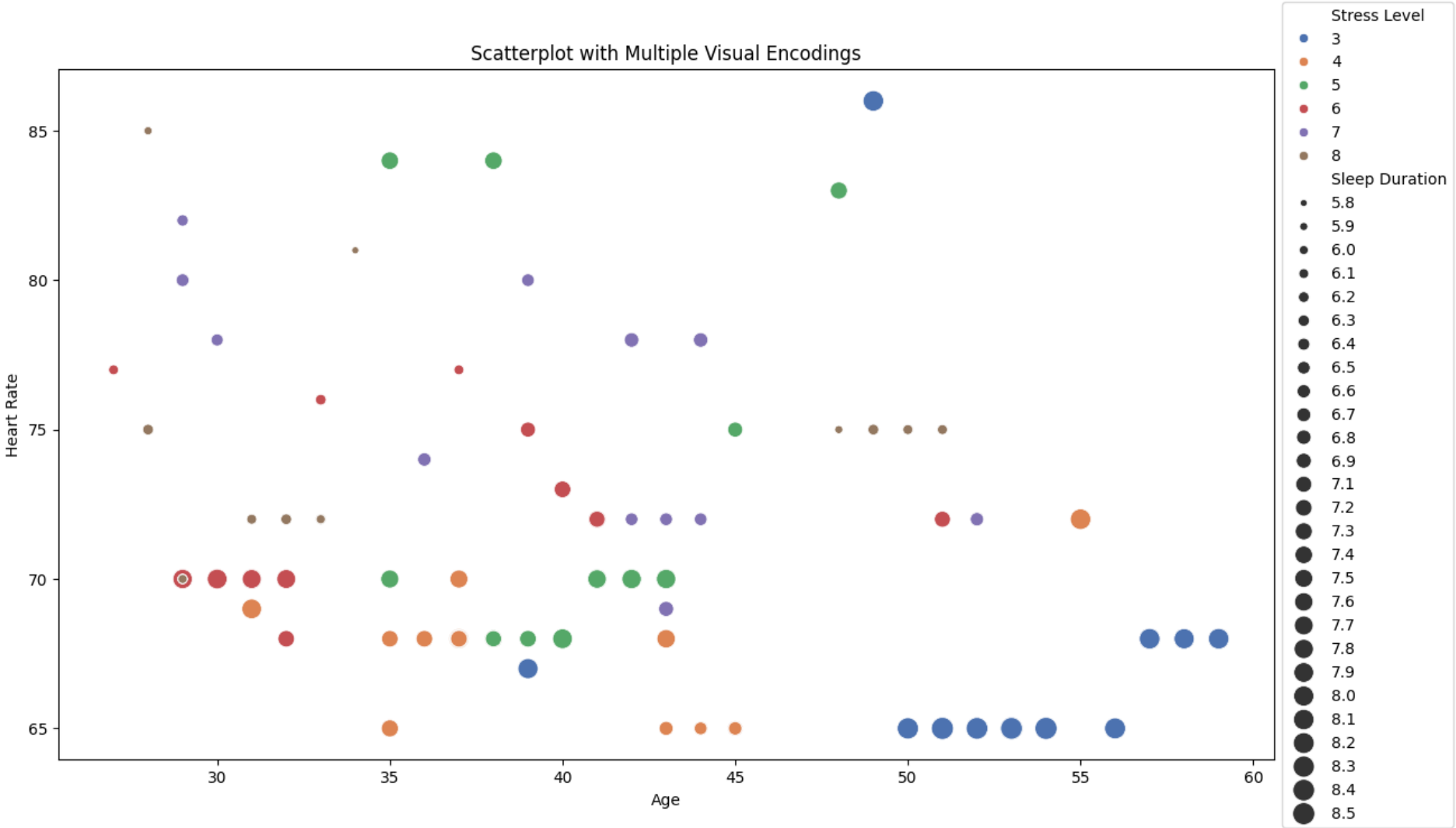
In [ ]:
```
sleep_disorder_count = df['Sleep Disorder'].value_counts()
colors = sns.color_palette("viridis", len(sleep_disorder_count))
fig, ax = plt.subplots(figsize=(8, 8))
ax.pie(sleep_disorder_count, labels=sleep_disorder_count.index, autopct='%1.1f%%', startangle=90, colors=colors)
ax.axis('equal')
plt.title('Distribution of Sleep Disorders')
plt.show()
```



The distribution highlights that while a substantial proportion of individuals report sleep disorders, a majority of the dataset does not exhibit significant sleep-related issues.

In [ ]:
```
plt.figure(figsize=(14, 8))
scatterplot=sns.scatterplot(x='Age', y='Heart Rate', hue='Stress Level', size='Sleep Duration', sizes=(20, 200), data=df, palette='deep', legend='full')
scatterplot.legend(loc='center left', bbox_to_anchor=(1, 0.5))
plt.title('Scatterplot with Multiple Visual Encodings')
```

```
plt.show()
```



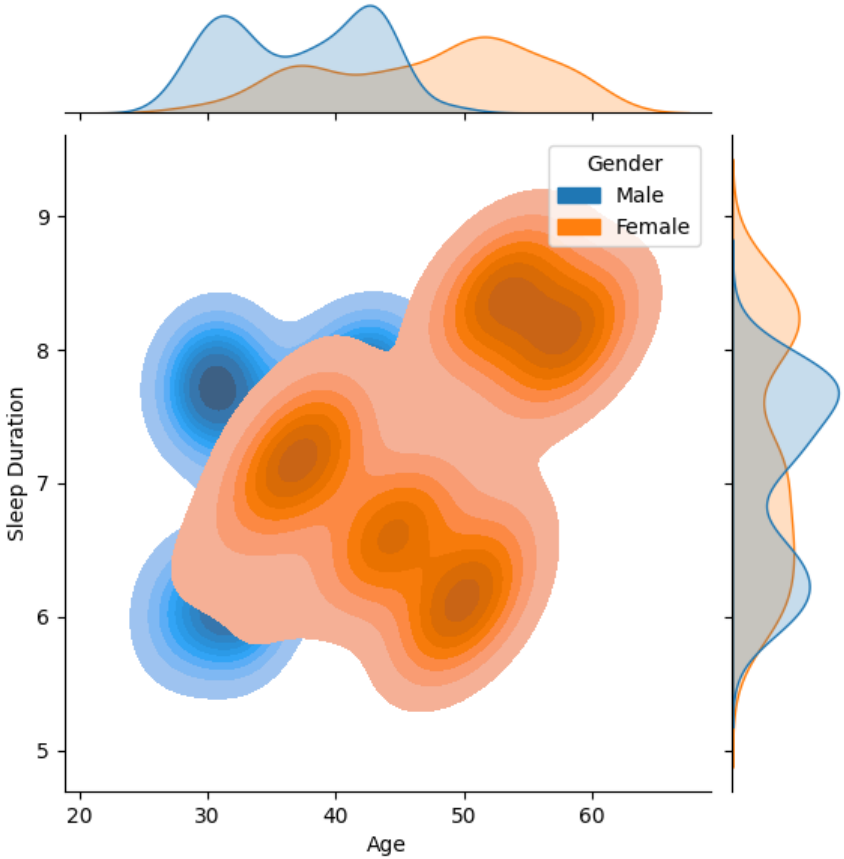
A noticeable trend emerges among individuals above the age of 50. These individuals tend to exhibit lower stress levels and enjoy a good amount of sleep duration. This could be indicative of a more relaxed lifestyle or reduced work-related stress in the later stages of life.

In the age range of 25 to 45, a cluster of points with heart rates between 68 and 75 stands out. Within this cluster, there is a pattern of higher stress levels and medium sleep duration. This suggests that individuals in this age range, with a specific heart rate range, tend to experience elevated stress levels despite achieving only moderate sleep durations.

In [ ]:
```
sns.jointplot(df, x='Age', y="Sleep Duration", kind='kde', hue='Gender', fill=True)
plt.title("Contour Plot Showcasing Sleep Duration of Different Age Group Filtering Genders", y=1.25)
```

```
plt.show()
```

Contour Plot Showcasing Sleep Duration of Different Age Group Filtering Genders

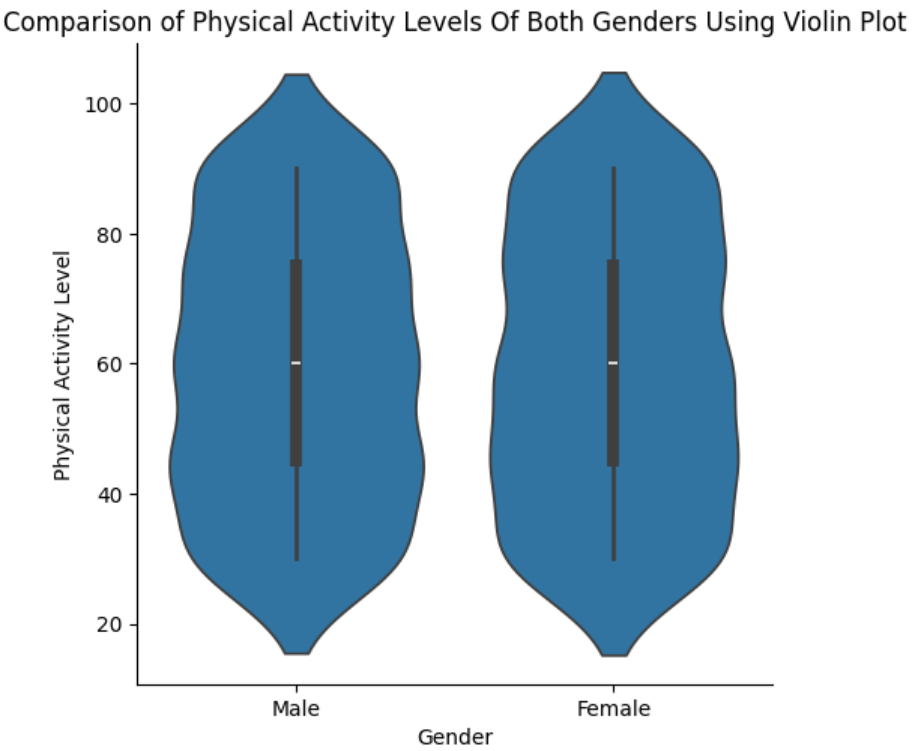


The data is filtered by gender, allowing for a comparison between male and female sleep duration patterns across different age groups.

A distinctive observation is the contour line for females above the age of 45. In this group, there is a minimum sleep duration of 6.5 hours. This suggests that a significant number of females in this age category tend to prioritize and achieve a minimum of 6.5 hours of sleep per night.This pattern might be indicative of health-conscious behavior or the recognition of the importance of sufficient sleep for well-being among females in this age group.

In contrast, the contour plot for males in the same age group does not exhibit a similar minimum sleep duration pattern. This could suggest that, on average, males in this age range may not prioritize a minimum of 6.5 hours of sleep as consistently as their female counterparts.

In [ ]:
```
sns.catplot(df, x ='Gender', y='Physical Activity Level', kind="violin")
plt.title("Comparison of Physical Activity Levels Of Both Genders Using Violin Plot")
plt.show()
```



The violin plot depicting physical activity levels for both genders reveals distinct patterns in the distribution of activity levels. While the overall shapes of the male and female violins are similar, the notable difference lies in the width of the violins. The narrower width of the female violin indicates a higher concentration of data points within a specific range of physical activity levels. This suggests that females in the dataset exhibit a more consistent pattern in their engagement with physical activity, with a majority falling within a narrower band.

The wider width of the male violin suggests a greater variability in physical activity levels among males. The broader distribution implies that males in the dataset demonstrate a more diverse range of activity levels, with some engaging in higher levels of physical activity while others exhibit lower levels.

Findings:

Key Conclusion

The comprehensive analysis of the dataset has some crucial aspects related to sleep health and lifestyle. Occupations such as Nursing, Teaching, and Sales emerge as potential risk factors for sleep disorders, necessitating targeted interventions in these professional sectors. Age-related patterns indicate that individuals above 50 tend to experience lower stress levels and maintain healthier sleep habits. Gender-specific variations highlight the significance of considering distinct sleep duration patterns in females above the age of 45. Moreover, the relationship between age, stress, sleep, and heart rate underscores the need for age-specific stress management interventions. The blood pressure distribution across genders emphasizes the importance of gender-specific considerations in cardiovascular health.

Trends & Insights

Our analysis has revealed important trends and insights. The robust positive correlation (0.88) between sleep duration and sleep quality underscores the interconnected nature of these crucial factors, suggesting that improvements in sleep duration may lead to enhanced sleep quality. Additionally, the moderate correlation (0.67) between stress levels and heart rates focuses on the link between stress and cardiovascular health, highlighting the need for comprehensive stress management strategies. The density plot mapping daily steps to BMI categories illuminates clear associations, providing actionable intelligence for targeted weight management processes. The identification of outliers and the observation of cluster overlaps within specific gender groups point to unique health patterns, prompting further exploration. Lastly, the observed trends in physical activity levels (55 - 85) with heart rates around 70 offer practical insights for promoting cardiovascular fitness within this specific range. These findings collectively contribute to a nuanced understanding of the complex bonding between sleep, stress, physical activity, and cardiovascular health, empowering our data visualizations with informed recommendations and personalized interventions to enhance overall well-being.

Suggestions/Recommendations

- Sleep Enhancement Strategies:

Encourage individuals to prioritize sufficient sleep duration, aiming for at least 7-8 hours per night, to potentially improve sleep quality. Provide education on sleep hygiene practices, such as maintaining a consistent sleep schedule

- Cardiovascular Fitness Promotion:

Promote cardiovascular fitness within the observed heart rate and physical activity range Encourage regular physical activity, such as brisk walking, jogging, or cycling, to enhance cardiovascular health. Develop fitness programs that align with individual fitness levels.

- Stress Management:

Implement comprehensive stress management programs that address both psychological and physiological aspects. Incorporate stress-reduction techniques such as mindfulness, meditation, and relaxation exercises to mitigate the impact of stress

- Corporate Wellness Programs:

Implement wellness programs in corporate settings that address the identified health factors. Offer workplace initiatives that promote healthy sleep habits, stress reduction activities, and opportunities for physical activity during breaks.

Data Visualizations Used

Box and Whisker Plot: Effective for illustrating the distribution, central tendency, and variability of sleep duration and physical activity levels across different age groups.

Line Plot: Used for visualizing the relationship between stress level and age. Each point on the line plot corresponds to a specific age and stress level, connected by a line to illustrate overall trend.

Correlation Matrix Heatmap: A comprehensive visualization to understand the relationships and strengths of correlations between different variables in your dataset.

Density Line Plot: Useful for highlighting the density and distribution of daily steps across different BMI categories, offering insights into the relationship between physical activity and weight status.

2D Distance Plot: Helpful for identifying outliers and assessing the distance between heart rates and physical activity levels, providing a nuanced understanding of unique health patterns, particularly between genders.

Histogram: Ideal for showing the frequency distribution of sleep disorders among individuals with different occupations, enabling a quick overview of the prevalence within each group.

Pie Chart: Offers a concise and visually appealing representation of the distribution of sleep disorders within the entire dataset.

Scatterplot: Provides a multidimensional view by incorporating multiple visual encodings, facilitating the exploration of relationships between stress level, sleep duration, age, and heart rate.

Contour Plot: Useful for showcasing the contours of sleep duration across different age groups, filtered by gender, allowing for a detailed visualization of sleep patterns.

Violin Plot: Offers insights into the distribution and density of physical activity levels among males and females, highlighting potential gender-specific patterns.